

Project 04

Due date: 04/17

Second Coding Update

Points: 150

In this part you should have your complete solution up and running, evaluated on the validation subset. Not yet perfect, but it should be ready for final testing. Usually, if you already implemented preprocessing, segmentation and feature extraction, the classification is left for this part.

What to do and deliver?

- A report (no page limit, but try to be concise; 1000-2000 words should suffice) as a separate "Part 4" section of the readme.md in your GitHub repository that includes:
 - A justification of the choice of classifier. For instance, if you selected SVM with RBF kernel, say why you think this classifier is good for your project. (30 points)
 - A classification accuracy achieved on the training and validation subsets. That is, how many objects were classified correctly and how many of them were classified incorrectly. If your project includes a detection task, report IoU metrics. You can choose to provide additional performance metrics: Precision-Recall, f-measure, ROC curves, etc. (20 points)
 - A short commentary related to the observed accuracy and ideas for improvements. For instance, if you see almost perfect accuracy on the training set, and much worse on the validation set, what does that mean? If this is not desired, what do you think you could do to improve the generalization capabilities of your solution? You can think about one really small improvement (out of all improvements you propose in this report) to be implemented before the final testing. (60 points).
 - Push your current code to your project repository (40 points). This code should implement what you described in the report. Provide instructions on how to run your code on a data example (attach this example to the code).

Note 1 (for teams): explain individual contributions of each team member (this is needed to have this assignment graded).

Note 2 (NNs): as we have now unlocked NNs in the course, feel free to include NN implementations in your project. If you opt to add NNs to your project, include comparisons to your original methods and explain your reasoning for such modifications.

Submission: PDF file with the GitHub link.